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**Bijan Davvaz, Soleha, Dian Winda
Setyawati, Imam Mukhlash & Rinurwati**

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Bijan Davvaz¹ · Soleha² · Dian Winda Setyawati² · Imam Mukhlash² · Rinurwati²

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Abstract

This paper is a continuation of ideas presented by Bagirmaz (Appl Algebra Eng Commun Comput 30(4):285–297, 2019). Indeed, we introduce the notion of near approximations in a ring, which is an extended notion of a rough approximations in a ring. Then we define lower and upper near subrings based on ideals in a ring and give some properties of such subrings. Furthermore, we obtain a comparison between these types of approximations and the approximations introduced by Davvaz (Inf Sci 176:2417–2437, 2006).

Keywords Near set · Lower and upper approximations · Rough set · Ring · Ideal

Mathematics Subject Classification 03E99 · 16Y99

1 Introduction

Rough set theory is introduced by Pawlak [21, 22] as a powerful mathematical tool for dealing with uncertainty of data. It relies only on the available data and attributes to work on the analysis of important features as well as generation of classification rules. The theory of rough set is an extension of set theory, in which a subset of a universe

✉ Bijan Davvaz
davvaz@yazd.ac.ir

Soleha
seha_07@matematika.its.ac.id

Dian Winda Setyawati
dian_ws_math@matematika.its.ac.id

Imam Mukhlash
imamm@matematika.its.ac.id

Rinurwati
rinur@matematika.its.ac.id

¹ Department of Mathematics, Yazd University, Yazd, Iran

² Department of Mathematics, Institut Teknologi Sepuluh Nopember, Kampus ITS, Sukolilo, Surabaya 60111, Indonesia

is described by a pair of ordinary sets called the lower and upper approximations. The starting point of rough set theory is that the elements of a universe having the same description are indiscernible with respect to the available information. The indiscernibility is characterized by an equivalence relation in the way that two elements are equivalent if and only if they are indiscernible from each other. As is well known, any equivalence relation defined on a universe $\mathcal{O}$ determines a partition of $\mathcal{O}$ into a collection of equivalence classes: each class contains all and only the elements that are mutually equivalent among them. Any partition of $\mathcal{O}$ represents a piece of knowledge about the elements of $\mathcal{O}$ which forms a classification, and so any equivalence class (i.e., elementary set) is interpreted as a granule of knowledge contained in the partition. In [10], Davvaz and Mahdavi pour introduced a generalization of standard rough set theory. This is done by introducing several new approximation operators that are based on the notion of covering is induced by the notion of a property on the universe of discourse.

Originally, the notion of nearness between sets (proximity relation between sets) was derived from a spatial meaning of distance by Efremovič [12]. Later, it would seem that the notion of proximity between sets was more abstract, i.e., proximity not limited to a spatial interpretation (see, also [13, 24]). Peters studied near sets in [23], where objects with affinities were considered perceptually near to each other, i.e., objects with similar descriptions to some degrees. Although any rough set has a non-empty boundary region, the near set may also be a null set. In fact, this is the main difference between a rough set and a near set. Peters and Wasilewski [25] introduced an approach to the foundations of information science considered in the context of near sets. Perceptual information systems (or, more concisely, perceptual systems) provide stepping stones leading to nearness relations, near sets and a framework for classifying perceptual objects.

Biswas and Nanda [3] introduced the notion of rough subgroups. Kuroki in [18], introduced the notion of a rough ideal in a semigroup. Kuroki and Wang [20] gave some properties of the lower and upper approximations with respect to the normal subgroups. Also, Kuroki and Mordeson in [19] studied the structure of rough sets and rough groups. Also, see [26]. Davvaz [4] concerned a relationship between rough sets and ring theory. He introduced the notion of rough subring (resp. ideal) with respect to an ideal of a ring which is an extended notion of a subring (resp. ideal) in a ring, and he gave some properties of the lower and the upper approximations in a ring. Davvaz [5] considered a ring as a universal set and assumed that the knowledge about objects is restricted by a fuzzy ideal. In fact, he applied the notion of fuzzy ideal of a ring for definitions of the lower and upper approximations in a ring. Davvaz and Mahdavi pour [9] considered an R -module as a universal set and introduced the notion of rough submodule with respect to a submodule of an R -module. Also, see [6–8, 11, 16, 17, 27, 28].

İnan and Öztürk [14, 15] investigated the basic concepts of the algebraic structures of the near set theory. They introduced the concept of near groups, weak cosets, near normal subgroups, homomorphism of near groups on nearness approximation spaces. Bağırmaz [2] introduced the notion of near subsemigroups, near ideals, near bi- ideals and homomorphisms of near semigroups on near approximation spaces. Recently, Bağırmaz [1] introduced the notion of a near approximations in a group, which is an extended notion of a rough approximations in a group. Then, he

studied lower and upper near subgroups based on normal subgroups in a group and gave some properties of such subgroups.

The paper is organized as follows: after the introduction, in Sect. 2 we review the fundamental definition of rough approximations, and in Sect. 3 we review some results about roughness in rings. In Sect. 4, we present some results about near sets and examples. In Sect. 5, investigate upper and lower near subrings and ideals and study their properties. In Sect. 6, compared the near approximations and the rough approximations in a ring.

2 Rough approximations

A pair $(\mathcal{O}, \rho)$ where $\mathcal{O} \neq \emptyset$ and ρ is an equivalence relation on $\mathcal{O}$, is called an approximation space. We denote the class of $x \in \mathcal{O}$ by $[x]_\rho$. For an approximation space $(\mathcal{O}, \rho)$, by a rough approximation in $(\mathcal{O}, \rho)$ we mean a mapping $Apr : \mathcal{P}(\mathcal{O}) \longrightarrow \mathcal{P}(\mathcal{O}) \times \mathcal{P}(\mathcal{O})$ defined by for every $A \in \mathcal{P}(\mathcal{O})$,

$$Apr_\rho(A) = (\underline{Apr}_\rho(A), \overline{Apr}_\rho(A)),$$

where

$$\underline{Apr}_\rho(A) = \{x \in \mathcal{O} : [x]_\rho \subseteq A\}$$

and

$$\overline{Apr}_\rho(A) = \{x \in \mathcal{O} : [x]_\rho \cap A \neq \emptyset\}.$$

$\underline{Apr}_\rho(A)$ is called a lower rough approximation of A in $(\mathcal{O}, \rho)$, where as $\overline{Apr}_\rho(A)$ is called upper rough approximation of $\mathcal{O}$ in $(\mathcal{O}, \rho)$.

Given an approximation space $(\mathcal{O}, \rho)$, a pair $(X, Y) \in \mathcal{P}(\mathcal{O}) \times \mathcal{P}(\mathcal{O})$ is called a rough set in $(\mathcal{O}, \rho)$ if and only if $(X, Y) = Apr_\rho(A)$ for some $A \in \mathcal{P}(\mathcal{O})$. A subset A of $\mathcal{O}$ is called definable if $\underline{Apr}_\rho(A) = \overline{Apr}_\rho(A)$. If $A \subseteq \mathcal{O}$ is given by a predicate P and $x \in \mathcal{O}$, then

1. $x \in \underline{Apr}_\rho(A)$ means that x certainly has property P ,
2. $x \in \overline{Apr}_\rho(A)$ means that x possibly has property P ,
3. $x \in \mathcal{O} \setminus \overline{Apr}_\rho(A)$ means that x definitely does not have property P .

3 Approximations in ring

A ring is an algebraic structure consisting of a set together with two binary operations usually called addition and multiplication, where the set is an abelian group under addition and a monoid under multiplication such that multiplication distributes over addition. In other words, a nonempty set R is said to be a ring if

in R there are defined two binary operations, denoted by $+$ and $\cdot$ respectively, such that for all a, b, c in R : (1) $a + b = b + a$, (2) $(a + b) + c = a + (b + c)$, (3) there is an element 0 in R such that $a + 0 = a$, (4) there exists an element $-a$ in R such that $a + (-a) = 0$, (5) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$, (6) $\cdot$ is distributive with respect to $+$, i.e., $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$. Axioms 1 through 4 merely state that R is an abelian group under the operation $+$. The additive identity of a ring is called the zero element. If $a \in R$ and $n \in \mathbb{Z}$, then na has its usual meaning for additive groups. If in multiplication: $a \cdot b = b \cdot a$ for all a, b in R , then R is said to be a commutative ring. If R contains an element 1 such that $1 \cdot a = a \cdot 1 = a$ for all a in R , then R is said to be a ring with unit element. A non-empty subset S of R is a subring of R if the following conditions hold: for every $a, b \in S$, $a - b \in S$ and $a \cdot b \in S$. A non-empty subset I of a ring R is said to be an ideal of R if (1) I is a subgroup of R under addition, (2) for every $a \in I$ and $r \in R$, both ar and ra are in I . Clearly, each ideal is a subring. If ρ is an equivalence relation on a ring R , then ρ is called a congruence relation if apb implies that $(a + x, b + x) \in \rho$, $(ax, bx) \in \rho$ and $(xa, xb) \in \rho$, for all $x \in R$. If I is an ideal of R , then induces a congruence relation ρ_I on R such that $(R/I, +, \cdot)$ is a ring. Lower and upper approximation operators of rough sets are applied into ring theory [4] and so the notion of a rough ring has been introduced. In [4], Davvaz introduced the concept of rough ring with respect to an ideal of a ring. Let R be a ring and I be an ideal of R . Let A be a non-empty subset of R . Then, the sets

$$\underline{Apr}_I(A) = \{x \in R : x + I \subseteq A\}$$

and

$$\overline{Apr}_I(A) = \{x \in R : (x + I) \cap A \neq \emptyset\},$$

are called, respectively, lower and upper approximations of the set A with respect to the ideal I .

Definition 3.1 [4] Let I be an ideal of R and $\underline{Apr}_I(A) = (\underline{Apr}_I(A), \overline{Apr}_I(A))$ a rough set in the approximation space (R, I) . If $\underline{Apr}_I(A)$ and $\overline{Apr}_I(A)$ are ideals (resp. subrings) of R , then we call $\underline{Apr}_I(A)$ a rough ideal (resp. subring). Note that a rough subring also is called a rough ring.

4 Near sets

In this section, we present the main definitions and properties of near sets as in [25].

Definition 4.1 [25] A perceptual information system $\langle \mathcal{O}, \mathbb{F} \rangle$ or, more concisely, perceptual system, is a real-valued total deterministic information system where $\mathcal{O}$ is a non-empty set of perceptual objects, while $\mathbb{F}$ is a countable set of probe functions.

An object description is defined by means of a tuple of function values $\Phi(x)$ associated with an object $x \in A$. The important thing to notice is the choice of

functions $\varphi_i \in \mathcal{B}$ used to describe an object of interest. Assume that $\mathcal{B} \subseteq \mathbb{F}$ (see Table 1) is a given set of functions representing features of sample objects $A \subseteq \mathcal{O}$.

Definition 4.2 [25] Let $\langle \mathcal{O}, \mathbb{F} \rangle$ be a perceptual system. For every $\mathcal{B} \subseteq \mathbb{F}$ the indiscernibility relation $\sim_{\mathcal{B}}$ is defined as follows:

$$\sim_{\mathcal{B}} = \{(x, y) \in \mathcal{O} \times \mathcal{O} : \forall \varphi_i \in \mathcal{B}, \varphi_i(x) = \varphi_i(y)\}.$$

If $\mathcal{B} = \{\varphi\}$, for some $\varphi \in \mathbb{F}$, instead of $\sim_{\{\varphi\}}$ we write $\sim_{\varphi}$.

This relation defines a partition of $\mathcal{O}$ into non-empty, pairwise disjoint subsets, where they are equivalence classes, and denoted by $[x]_{\mathcal{B}}$, where $[x]_{\mathcal{B}} = \{y \in \mathcal{O} : x \sim_{\mathcal{B}} y\}$. These classes form the quotient set $\mathcal{O} / \sim_{\mathcal{B}}$, where $\mathcal{O} / \sim_{\mathcal{B}} = \{[x]_{\mathcal{B}} : x \in \mathcal{O}\}$.

Example 1 Let $\langle \mathcal{O}, \mathbb{F} \rangle$ be a perceptual system, where $\mathcal{O} = \{x_i : 1 \leq i \leq 10\}$ and $\mathbb{F} = \{\varphi_1, \varphi_2, \varphi_3\}$ and the values of probe functions from $\mathbb{F}$ to percepts are defined in Table 2. Then, the partitions are as follows:

$$\begin{aligned} \mathcal{O} / \sim_{\varphi_1} &= \left\{ \{x_1, x_3, x_4, x_6, x_9, x_{10}\}, \{x_2, x_5, x_7, x_8\} \right\}, \\ \mathcal{O} / \sim_{\varphi_2} &= \left\{ \{x_1, x_5, x_7, x_8\}, \{x_2, x_6, x_{10}\}, \{x_3, x_4, x_9\} \right\}, \\ \mathcal{O} / \sim_{\varphi_3} &= \left\{ \{x_1, x_3, x_5, x_6, x_9\}, \{x_2, x_4, x_{10}\}, \{x_7, x_8\} \right\}. \end{aligned}$$

Example 2 Consider Example 1 and suppose that $\mathcal{B} = \{\varphi_1, \varphi_2\}$ and $\mathcal{B}' = \{\varphi_1, \varphi_2, \varphi_3\}$. Then, we have

$$\begin{aligned} \mathcal{O} / \sim_{\mathcal{B}} &= \left\{ \{x_1\}, \{x_2\}, \{x_3, x_4, x_9\}, \{x_5, x_7, x_8\}, \{x_6, x_{10}\} \right\}, \\ \mathcal{O} / \sim_{\mathcal{B}'} &= \left\{ \{x_1\}, \{x_2\}, \{x_3, x_9\}, \{x_4\}, \{x_5\}, \{x_6\}, \{x_7, x_8\}, \{x_{10}\} \right\}. \end{aligned}$$

The basic idea in the near set approach to object recognition is to compare object descriptions. Sample perceptual objects $x, y \in \mathcal{O}$ ($x \neq y$) are near each other if and

Table 1 Description symbols

Symbol	Interpretation
$\mathbb{R}$	Set of real numbers
$\mathcal{O}$	Set of perceptual objects
A	$A \subseteq \mathcal{O}$, set of sample objects
x	$x \in \mathcal{O}$, sample object
$\mathbb{F}$	A set of functions representing object features
$\mathcal{B}$	$\mathcal{B} \subseteq \mathbb{F}$
Φ	$\Phi : \mathcal{O} \rightarrow \mathbb{R}^n$, object description
n	n is a description length
φ_i	$\varphi_i \in \mathcal{B}$, where $\varphi_i : \mathcal{O} \rightarrow \mathbb{R}$
$\Phi(x)$	$\Phi(x) = (\varphi_1(x), \dots, \varphi_n(x))$, description

Table 2 Perceptual information system of Example 1

	φ_1	φ_2	φ_3
x_1	1	0	3
x_2	4	2	6
x_3	1	5	3
x_4	1	5	6
x_5	4	0	3
x_6	1	2	3
x_7	4	0	7
x_8	4	0	7
x_9	1	5	3
x_{10}	1	2	6

only if x and y have similar descriptions. Let A and A' are two subsets of $\mathcal{O}$ and $\mathcal{B} \subseteq \mathbb{F}$. We say that A is near A' if there exist $x \in A$, $y \in A'$ and $\varphi \in \mathcal{B}$ such that $x \sim_{\varphi} y$.

Definition 4.3 The family of neighborhood with respect to one object feature is defined as:

$$N_1(\mathcal{B}) = \cup\{[x]_{\varphi_i} : x \in \mathcal{O}, \varphi_i \in \mathcal{B}\}.$$

The family of neighborhoods with respect to two object features is defined as

$$N_2(\mathcal{B}) = \cup\{[x]_{\{\varphi_i, \varphi_j\}} : x \in \mathcal{O}, \varphi_i, \varphi_j \in \mathcal{B}, i \neq j\}.$$

Finally, the family of neighborhoods with respect to all object features is defined as

$$N_{|\mathcal{B}|}(\mathcal{B}) = \cup\{[x]_{\{\varphi_1, \dots, \varphi_{|\mathcal{B}|}\}} : x \in \mathcal{O}, \varphi_i \in \mathcal{B}\}.$$

Example 3 Let $\langle \mathcal{O}, \mathbb{F} \rangle$ be a perceptual system, where $\mathcal{O} = \{x_i : 1 \leq i \leq 12\}$ and $\mathcal{B} = \{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$ and the values of probe functions from $\mathbb{F}$ to percepts are defined in Table 3.

$$\begin{aligned} \mathcal{O} / \sim_{\varphi_1} &= \left\{ \{x_1, x_3, x_7, x_{10}, x_{11}, x_{12}\}, \{x_2, x_4, x_5, x_6, x_8, x_9\} \right\} = \left\{ [x_1]_{\varphi_1}, [x_2]_{\varphi_1} \right\}, \\ \mathcal{O} / \sim_{\varphi_2} &= \left\{ \{x_1, x_5, x_{12}\}, \{x_2, x_6, x_7, x_8\}, \{x_3, x_4, x_9, x_{10}, x_{11}\} \right\} \\ &= \left\{ [x_1]_{\varphi_2}, [x_2]_{\varphi_2}, [x_3]_{\varphi_2} \right\}, \\ \mathcal{O} / \sim_{\varphi_3} &= \left\{ \{x_1, x_5, x_7, x_{12}\}, \{x_2, x_3, x_4, x_6, x_8, x_9, x_{10}, x_{11}\} \right\} = \left\{ [x_1]_{\varphi_3}, [x_2]_{\varphi_3} \right\}, \\ \mathcal{O} / \sim_{\varphi_4} &= \left\{ \{x_1, x_3, x_{10}, x_{11}, x_{12}\}, \{x_2, x_4, x_6, x_8\}, \{x_5, x_7, x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_4}, [x_2]_{\varphi_4}, [x_5]_{\varphi_4} \right\}. \end{aligned}$$

Table 3 Perceptual information system of Example 3

	φ_1	φ_2	φ_3	φ_4
x_1	0	1	5	7
x_2	3	2	6	8
x_3	0	4	6	7
x_4	3	4	6	8
x_5	3	1	5	9
x_6	3	2	6	8
x_7	0	2	5	9
x_8	3	2	6	8
x_9	3	4	6	9
x_{10}	0	4	6	7
x_{11}	0	4	6	7
x_{12}	0	1	5	7

Hence, we have

$$N_1(\mathcal{B}) = \left\{ [x_1]_{\varphi_1}, [x_2]_{\varphi_1}, [x_1]_{\varphi_2}, [x_2]_{\varphi_2}, [x_3]_{\varphi_2}, [x_1]_{\varphi_3}, [x_2]_{\varphi_3}, [x_1]_{\varphi_4}, [x_2]_{\varphi_4}, [x_5]_{\varphi_4} \right\}.$$

In continue we use the following abbreviated notation: the set $\{\varphi_i, \dots, \varphi_j\}$ will be denoted by $\varphi_{i\dots j}$. Now, we obtain the equivalence classes of two features:

$$\begin{aligned} \mathcal{O} / \sim_{\varphi_{12}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4, x_9\}, \{x_5\}, \{x_7\} \right\} \\ &= \left\{ [x_1]_{\varphi_{12}}, [x_2]_{\varphi_{12}}, [x_3]_{\varphi_{12}}, [x_4]_{\varphi_{12}}, [x_5]_{\varphi_{12}}, [x_7]_{\varphi_{12}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{13}} &= \left\{ \{x_1, x_7, x_{12}\}, \{x_2, x_4, x_6, x_8, x_9\}, \{x_3, x_{10}, x_{11}\}, \{x_5\} \right\} \\ &= \left\{ [x_1]_{\varphi_{13}}, [x_2]_{\varphi_{13}}, [x_3]_{\varphi_{13}}, [x_5]_{\varphi_{13}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{14}} &= \left\{ \{x_1, x_3, x_{10}, x_{11}, x_{12}\}, \{x_2, x_4, x_6, x_8\}, \{x_5, x_9\}, \{x_7\} \right\} \\ &= \left\{ [x_1]_{\varphi_{14}}, [x_2]_{\varphi_{14}}, [x_3]_{\varphi_{14}}, [x_7]_{\varphi_{14}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{23}} &= \left\{ \{x_1, x_5, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_4, x_9, x_{10}, x_{11}\}, \{x_7\} \right\} \\ &= \left\{ [x_1]_{\varphi_{23}}, [x_2]_{\varphi_{23}}, [x_3]_{\varphi_{23}}, [x_7]_{\varphi_{23}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{24}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4\}, \{x_5\}, \{x_7\}, \{x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_{24}}, [x_2]_{\varphi_{24}}, [x_3]_{\varphi_{24}}, [x_4]_{\varphi_{24}}, [x_5]_{\varphi_{24}}, [x_7]_{\varphi_{24}}, [x_9]_{\varphi_{24}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{34}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_4, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_5, x_7\}, \{x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_{34}}, [x_2]_{\varphi_{34}}, [x_3]_{\varphi_{34}}, [x_5]_{\varphi_{34}}, [x_9]_{\varphi_{34}} \right\}. \end{aligned}$$

Thus, the family of neighborhoods with respect to two object features is

$$N_2(\mathcal{B}) = \left\{ [x_1]_{\varphi_{12}}, [x_2]_{\varphi_{12}}, [x_3]_{\varphi_{12}}, [x_4]_{\varphi_{12}}, [x_5]_{\varphi_{12}}, [x_7]_{\varphi_{12}}, [x_1]_{\varphi_{13}}, [x_2]_{\varphi_{13}}, [x_3]_{\varphi_{13}}, \right. \\ [x_5]_{\varphi_{13}}, [x_1]_{\varphi_{13}}, [x_2]_{\varphi_{13}}, [x_3]_{\varphi_{13}}, [x_5]_{\varphi_{13}}, [x_1]_{\varphi_{23}}, [x_2]_{\varphi_{23}}, [x_3]_{\varphi_{23}}, [x_7]_{\varphi_{23}}, \\ [x_1]_{\varphi_{24}}, [x_2]_{\varphi_{24}}, [x_3]_{\varphi_{24}}, [x_4]_{\varphi_{24}}, [x_5]_{\varphi_{24}}, [x_7]_{\varphi_{24}}, [x_9]_{\varphi_{24}}, [x_1]_{\varphi_{34}}, [x_2]_{\varphi_{34}}, \\ \left. [x_3]_{\varphi_{34}}, [x_5]_{\varphi_{34}}, [x_9]_{\varphi_{34}} \right\}.$$

Now, we obtain the equivalence classes of three features:

$$\begin{aligned} \mathcal{O} / \sim_{\varphi_{123}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4, x_9\}, \{x_5\}, \{x_7\} \right\} \\ &= \left\{ [x_1]_{\varphi_{123}}, [x_2]_{\varphi_{123}}, [x_3]_{\varphi_{123}}, [x_4]_{\varphi_{123}}, [x_5]_{\varphi_{123}}, [x_7]_{\varphi_{123}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{134}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_4, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_5\}, \{x_7\}, \{x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_{134}}, [x_2]_{\varphi_{134}}, [x_3]_{\varphi_{134}}, [x_5]_{\varphi_{134}}, [x_7]_{\varphi_{134}}, [x_9]_{\varphi_{134}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{124}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4\}, \{x_5\}, \{x_7\}, \{x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_{124}}, [x_2]_{\varphi_{124}}, [x_3]_{\varphi_{124}}, [x_4]_{\varphi_{124}}, [x_5]_{\varphi_{124}}, [x_7]_{\varphi_{124}}, [x_9]_{\varphi_{124}} \right\}, \\ \mathcal{O} / \sim_{\varphi_{234}} &= \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4\}, \{x_5\}, \{x_7\}, \{x_9\} \right\} \\ &= \left\{ [x_1]_{\varphi_{234}}, [x_2]_{\varphi_{234}}, [x_3]_{\varphi_{234}}, [x_4]_{\varphi_{234}}, [x_5]_{\varphi_{234}}, [x_7]_{\varphi_{234}}, [x_9]_{\varphi_{234}} \right\}. \end{aligned}$$

Hence, the family of neighborhoods with respect to three object features is

$$N_3(\mathcal{B}) = \left\{ [x_1]_{\varphi_{123}}, [x_2]_{\varphi_{123}}, [x_3]_{\varphi_{123}}, [x_4]_{\varphi_{123}}, [x_5]_{\varphi_{123}}, [x_7]_{\varphi_{123}}, [x_1]_{\varphi_{134}}, [x_2]_{\varphi_{134}}, \right. \\ [x_3]_{\varphi_{134}}, [x_5]_{\varphi_{134}}, [x_7]_{\varphi_{134}}, [x_9]_{\varphi_{134}}, [x_1]_{\varphi_{124}}, [x_2]_{\varphi_{124}}, [x_3]_{\varphi_{124}}, [x_4]_{\varphi_{124}}, [x_5]_{\varphi_{124}}, \\ [x_7]_{\varphi_{124}}, [x_9]_{\varphi_{124}}, [x_1]_{\varphi_{234}}, [x_2]_{\varphi_{234}}, [x_3]_{\varphi_{234}}, [x_4]_{\varphi_{234}}, [x_5]_{\varphi_{234}}, [x_7]_{\varphi_{234}}, \\ \left. [x_9]_{\varphi_{234}} \right\}.$$

Finally, we have

$$\mathcal{O} / \sim_B = \left\{ \{x_1, x_{12}\}, \{x_2, x_6, x_8\}, \{x_3, x_{10}, x_{11}\}, \{x_4\}, \{x_5\}, \{x_7\}, \{x_9\} \right\},$$

and so

$$N_4(\mathcal{B}) = \left\{ [x_1]_B, [x_2]_B, [x_3]_B, [x_4]_B, [x_5]_B, [x_7]_B, [x_9]_B \right\}.$$

Definition 4.4 [23] A nearness approximation space (NAS) is a tuple $NAS = (\mathcal{O}, \mathbb{F}, \sim_B, \mathcal{B}_r, N_r)$ where the approximation space NAS is defined with a set of perceived objects $\mathcal{O}$, set of probe functions $\mathbb{F}$ representing object features,

indiscernibility relation $\sim_{\mathcal{B}_r}$ defined relative to $\mathcal{B}_r \subseteq \mathcal{B} \subseteq \mathbb{F}$, collection of partitions (families of neighbourhoods) $N_r(\mathcal{B})$.

Definition 4.5 [23] Let $(\mathcal{O}, \mathbb{F}, \sim_{\mathcal{B}_r}, \mathcal{B}_r, N_r)$ be a nearness approximation space and let A be an arbitrary subset of $\mathcal{O}$. We define

$$\underline{N_r(\mathcal{B})}(A) = \{x \in \mathcal{O} : [x]_{\mathcal{B}_r} \subseteq A\}$$

and

$$\overline{N_r(\mathcal{B})}(A) = \{x \in \mathcal{O} : [x]_{\mathcal{B}_r} \cap A \neq \emptyset\}.$$

$\underline{N_r(\mathcal{B})}(A)$ is called $N_r(\mathcal{B})$ -lower approximation and $\overline{N_r(\mathcal{B})}(A)$ is called $N_r(\mathcal{B})$ -upper approximation of A in $(\mathcal{O}, \mathbb{F}, \sim_{\mathcal{B}_r}, \mathcal{B}_r, N_r)$.

Example 4 In Example 3, suppose that $A = \{x_1, x_2, x_5, x_7, x_9, x_{12}\}$. Then, we have

$$\begin{aligned}\underline{N_1(\mathcal{B})}(A) &= \{x_1, x_5, x_7, x_9, x_{12}\}, \\ \overline{N_1(\mathcal{B})}(A) &= \mathcal{O}, \\ \underline{N_4(\mathcal{B})}(A) &= \{x_1, x_5, x_7, x_9, x_{12}\}, \\ \overline{N_4(\mathcal{B})}(A) &= \{x_1, x_2, x_5, x_6, x_7, x_8, x_9, x_{12}\}.\end{aligned}$$

5 Near approximation in a ring

We denote $\mathbb{F}$ the set of all ideals in a ring R and $\mathcal{B} \subseteq \mathbb{F}$. In this case, we use the pair $(R, \mathbb{F})$ instead of the nearness approximation space $(\mathcal{O}, \mathbb{F}, \sim_{\mathcal{B}_r}, \mathcal{B}_r, N_r)$.

If $(R, \mathbb{F})$ is a nearness approximation space, then

$$[x]_{\mathcal{B}_r} = x + \mathcal{B}_r = x + \left(\bigcap_{I \in \mathcal{B}_r} I \right).$$

Definition 5.1 Let $(R, \mathbb{F})$ be a nearness approximation space and A be a non-empty subset of R . Then, the sets

$$\underline{N_r(\mathcal{B})}(A) = \{x \in R : x + \mathcal{B}_r \subseteq A\}$$

and

$$\overline{N_r(\mathcal{B})}(A) = \{x \in R : (x + \mathcal{B}_r) \cap A \neq \emptyset\}.$$

are called near lower and near upper approximations of A with respect to the ideals of R , respectively.

Proposition 5.2 For every approximation space $(R, \mathbb{F})$ and every subsets $A, B \subseteq R$, we have

1. $\underline{N_r(\mathcal{B})}(A) \subseteq A \subseteq \overline{N_r(\mathcal{B})}(A)$;
2. $\underline{N_r(\mathcal{B})}(\emptyset) = \emptyset = \overline{N_r(\mathcal{B})}(\emptyset)$;
3. $\underline{N_r(\mathcal{B})}(R) = R = \overline{N_r(\mathcal{B})}(R)$;
4. If $A \subseteq B$, then $\underline{N_r(\mathcal{B})}(A) \subseteq \underline{N_r(\mathcal{B})}(B)$ and $\overline{N_r(\mathcal{B})}(A) \subseteq \overline{N_r(\mathcal{B})}(B)$;
5. $\underline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A)) = \underline{N_r(\mathcal{B})}(A)$;
6. $\overline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A)) \supseteq \overline{N_r(\mathcal{B})}(A)$;
7. $\overline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A)) \supseteq \underline{N_r(\mathcal{B})}(A)$;
8. $\underline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A)) = \overline{N_r(\mathcal{B})}(A)$;
9. $\underline{N_r(\mathcal{B})}(A) \supseteq \overline{N_r(\mathcal{B})}(A^C)^C$;
10. $\overline{N_r(\mathcal{B})}(A) \supseteq \underline{N_r(\mathcal{B})}(A^C)^C$;
11. $\underline{N_r(\mathcal{B})}(A \cap B) = \underline{N_r(\mathcal{B})}(A) \cap \underline{N_r(\mathcal{B})}(B)$;
12. $\overline{N_r(\mathcal{B})}(A \cap B) \subseteq \overline{N_r(\mathcal{B})}(A) \cap \overline{N_r(\mathcal{B})}(B)$;
13. $\underline{N_r(\mathcal{B})}(A \cup B) \supseteq \underline{N_r(\mathcal{B})}(A) \cup \underline{N_r(\mathcal{B})}(B)$;
14. $\overline{N_r(\mathcal{B})}(A \cup B) = \overline{N_r(\mathcal{B})}(A) \cup \overline{N_r(\mathcal{B})}(B)$.

Proof (1), (4), (11), (12), (13) and (14): Since each ideal is a normal subgroup, the proof results from Proposition 3 in [1]

(2) and (3): The proof results from Definition 5.1.

(5): By part (1), we have $\underline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A)) \subseteq \underline{N_r(\mathcal{B})}(A)$. Conversely, let x be any element of $\underline{N_r(\mathcal{B})}(A)$. Then $x + \mathcal{B}_r \subseteq A$ for some $\mathcal{B}_r \subseteq \mathcal{B}$ so that $x + \mathcal{B}_r \subseteq \underline{N_r(\mathcal{B})}(A)$ for some $\mathcal{B}_r \subseteq \mathcal{B}$ which implies that $x \in \underline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A))$. Therefore, we obtain $\underline{N_r(\mathcal{B})}(A) \subseteq \underline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A))$.

(6): By part (1), we have $\overline{N_r(\mathcal{B})}(A) \subseteq \overline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A))$.

(7): By part (1), we have $\underline{N_r(\mathcal{B})}(A) \subseteq \underline{N_r(\mathcal{B})}(\underline{N_r(\mathcal{B})}(A))$.

(8): By part (1), we have $\underline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A)) \subseteq \overline{N_r(\mathcal{B})}(A)$. Conversely, let x be any element of $\overline{N_r(\mathcal{B})}(A)$. Then $x + \mathcal{B}_r \cap A \neq \emptyset$ for some $\mathcal{B}_r \subseteq \mathcal{B}$ so that $x + \mathcal{B}_r \subseteq \overline{N_r(\mathcal{B})}(A)$ for some $\mathcal{B}_r \subseteq \mathcal{B}$ which implies $x \in \underline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A))$. Therefore, $\overline{N_r(\mathcal{B})}(A) \subseteq \underline{N_r(\mathcal{B})}(\overline{N_r(\mathcal{B})}(A))$.

(9): Let x be any element of $\left(\overline{N_r(\mathcal{B})}(A^C)\right)^C$, then $x \notin \overline{N_r(\mathcal{B})}(A^C)$ so that $x + \mathcal{B}_r \cap A^C = \emptyset$ for every $\mathcal{B}_r \subseteq \mathcal{B}$. Hence, we have $x + \mathcal{B}_r \subseteq A$ for every $\mathcal{B}_r \subseteq \mathcal{B}$, which implies $x \in \underline{N_r(\mathcal{B})}(A)$. Therefore, $\left(\overline{N_r(\mathcal{B})}(A^C)\right)^C \subseteq \underline{N_r(\mathcal{B})}(A)$.

(10): Let x be any element $\left(\overline{N_r(\mathcal{B})(A^C)}\right)^C$, then $x \notin \overline{N_r(\mathcal{B})(A^C)}$ so that $x + \mathcal{B}_r \not\subseteq A^C$ for every $\mathcal{B}_r \subseteq \mathcal{B}$. Hence, we have $x + \mathcal{B}_r \cap A \neq \emptyset$ for every $\mathcal{B}_r \subseteq \mathcal{B}$, which implies $x \in \overline{N_r(\mathcal{B})(A)}$. Therefore, $\left(\overline{N_r(\mathcal{B})(A^C)}\right)^C \subseteq \overline{N_r(\mathcal{B})(A)}$. $\square$

The following examples show that the equality in (6), (7), (9) and (10) in Proposition 5.2 is not true in general.

Example 5 Let $R = \mathbb{Z}_{12}$, $\mathcal{B} = \{I_1, I_2\}$, $I_1 = \langle 4 \rangle = \{0, 4, 8\}$, $I_2 = \langle 6 \rangle = \{0, 6\}$ and $A = \{0, 2, 4, 6, 10, 11\}$ then

$$\begin{aligned} R/I_1 &= \{\{0, 4, 8\}, \{1, 5, 9\}, \{2, 6, 10\}, \{3, 7, 11\}\}, \\ R/I_2 &= \{\{0, 6\}, \{1, 7\}, \{2, 8\}, \{3, 9\}, \{4, 10\}, \{5, 11\}\}, \\ \overline{N_1(\mathcal{B})(A)} &= \{2, 6, 10\} \cup \{0, 6\} \cup \{4, 10\} = \{0, 2, 4, 6, 10\}, \\ \overline{N_1(\mathcal{B})(A)} &= \{0, 4, 8\} \cup \{2, 6, 10\} \cup \{3, 7, 11\} \cup \{0, 6\} \cup \{2, 8\} \cup \{4, 10\} \cup \{5, 11\} \\ &= \mathbb{Z}_{12} - \{1, 9\}, \\ \overline{N_1(\mathcal{B})(N_1(\mathcal{B})(A))} &= \{2, 6, 10\} \cup \{0, 6\} \cup \{4, 10\} = \{0, 2, 4, 6, 10\}, \\ \overline{N_1(\mathcal{B})(\overline{N_1(\mathcal{B})(A)})} &= \{0, 4, 8\} \cup \{2, 6, 10\} \cup \{3, 7, 11\} \cup \{0, 6\} \cup \{2, 8\} \cup \{4, 10\} \cup \{5, 11\} \\ &= \mathbb{Z}_{12} - \{1, 9\}, \\ \overline{N_1(\mathcal{B})(N_1(\mathcal{B})(A))} &= \{0, 4, 8\} \cup \{2, 6, 10\} \cup \{0, 6\}, \{2, 8\} \cup \{4, 10\} = \{0, 2, 4, 6, 8, 10\}, \\ \overline{N_1(\mathcal{B})(\overline{N_1(\mathcal{B})(A)})} &= \mathbb{Z}_{12}. \end{aligned}$$

This shows that

$$\begin{aligned} \overline{N_1(\mathcal{B})(N_1(\mathcal{B})(A))} &= \overline{N_1(\mathcal{B})(A)}, \\ \overline{N_1(\mathcal{B})(\overline{N_1(\mathcal{B})(A)})} &= \overline{N_1(\mathcal{B})(A)}, \\ \overline{N_1(\mathcal{B})(N_1(\mathcal{B})(A))} &\supset \overline{N_1(\mathcal{B})(A)}, \\ \overline{N_1(\mathcal{B})(\overline{N_1(\mathcal{B})(A)})} &\supset \overline{N_1(\mathcal{B})(A)}. \end{aligned}$$

Example 6 Let $R = \mathbb{Z}_{12}$, $\mathcal{B} = \{I_1, I_2\}$, $I_1 = \langle 4 \rangle = \{0, 4, 8\}$, $I_2 = \langle 6 \rangle = \{0, 6\}$ and $A = \{0, 3, 6, 8, 9\}$ and $A^C = \{1, 2, 4, 5, 7, 10, 11\}$ then

$$\begin{aligned} R/I_1 &= \{\{0, 4, 8\}, \{1, 5, 9\}, \{2, 6, 10\}, \{3, 7, 11\}\}, \\ R/I_2 &= \{\{0, 6\}, \{1, 7\}, \{2, 8\}, \{3, 9\}, \{4, 10\}, \{5, 11\}\}, \\ \overline{N_1(\mathcal{B})(A)} &= \{0, 6\} \cup \{3, 9\} = \{0, 3, 6, 9\}, \\ \overline{N_1(\mathcal{B})(A)} &= \mathbb{Z}_{12}, \\ \overline{N_1(\mathcal{B})(A^C)} &= \{1, 7\} \cup \{4, 10\} \cup \{5, 11\} = \{1, 4, 5, 7, 10, 11\}, \\ \overline{N_1(\mathcal{B})(A^C)} &= \mathbb{Z}_{12}. \end{aligned}$$

This shows that

$$\begin{aligned}\left(\underline{N_1(\mathcal{B})}(A^C)\right)^C &= \{0, 2, 3, 6, 8, 9\} \subset \overline{N_1(\mathcal{B})}(A), \\ \left(\overline{N_1(\mathcal{B})}(A^C)\right)^C &= \emptyset \subset \underline{N_1(\mathcal{B})}(A)\end{aligned}$$

In Example 7, three different approximations of a subset A of R are defined by ideals I_1, I_2 and I_3 .

Example 7 Let $R = \left(\left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{Z}_{18}\right\}, \oplus, \otimes\right)$ and $\mathcal{B} = \{I_1, I_2, I_3\}$, where

$$\begin{aligned}I_1 &= \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \{0, 9\}, \text{ and } b, c \in \mathbb{Z}_{18}\right\}, \\ I_2 &= \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \{0, 6, 12\}, \text{ and } b, c \in \mathbb{Z}_{18}\right\}, \\ I_3 &= \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \{0, 3, 6, 9, 12, 15\}, \text{ and } b, c \in \mathbb{Z}_{18}\right\}.\end{aligned}$$

Then, I_1, I_2 and I_3 are ideals of R .

Let $A = \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \{0, 1, 3, 5, 6, 9, 11, 13, 15\} \text{ and } b, c \in \mathbb{Z}_{18}\right\}$. For $r = 1$, the following cosets are in $N_1(\mathcal{B})$,

$$\begin{aligned}R/I_1 &= \left\{\left[\begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_1} : k \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}\right\}, \\ R/I_2 &= \left\{\left[\begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_2} : k \in \{0, 1, 2, 3, 4, 5\}\right\}, \\ R/I_3 &= \left\{\left[\begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_3} : k \in \{0, 1, 2\}\right\}.\end{aligned}$$

Then, by a calculation we get

$$\begin{aligned}N_1(\mathcal{B}) &= \cup\{R/\mathcal{B}_1 : \mathcal{B}_1 \subseteq \mathcal{B}\} = \{R/I_1, R/I_2, R/I_3\}, \\ \underline{N_1(\mathcal{B})}(A) &= \cup\{x + \mathcal{B}_1 : x + \mathcal{B}_1 \subseteq A\} \\ &= \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_1} \cup \left[\begin{pmatrix} 6 & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_1} \cup \left[\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}\right]_{I_2} \\ &= \left\{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \{0, 3, 6, 9, 15\}, \text{ and } b, c \in \mathbb{Z}_{18}\right\},\end{aligned}$$

and

$$\begin{aligned}
 \overline{N_1(\mathcal{B})}(A) &= \cup \{x + \mathcal{B}_1 : x + \mathcal{B}_1 \cap A \neq \emptyset\} \\
 &= \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \cup \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \cup \left[\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \\
 &\cup \left[\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \cup \left[\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \cup \left[\begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \\
 &\cup \left[\begin{pmatrix} 6 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_1} \cup \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_2} \cup \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_2} \\
 &\cup \left[\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_2} \cup \left[\begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_2} \cup \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_3} \\
 &\cup \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_3} \cup \left[\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \right]_{I_3} \\
 &= R.
 \end{aligned}$$

For $r = 2$, the subsets $\mathcal{B}_2 \subseteq \mathcal{B}$ are $\{I_1, I_2\}, \{I_1, I_3\}, \{I_2, I_3\}$. By the above explanation, we have $[x]_{\{I_1, I_2\}} = x + (I_1 \cap I_2)$. Then, by a simple calculation we have

$$\begin{aligned}
 \overline{N_2(\mathcal{B})}(A) &= A, \\
 \overline{N_2(\mathcal{B})}(A) &= \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a \in \mathbb{Z}_{18}/\{8, 16\} \text{ and } b, c \in \mathbb{Z}_{18} \right\}.
 \end{aligned}$$

For $r = 3$, by the above property, we obtain $\overline{[x]_{\{I_1, I_2, I_3\}}} = x + (I_1 \cap I_2 \cap I_3)$. Then, by a calculation we obtain $\overline{N_3(\mathcal{B})}(A) = A$ and $\overline{N_3(\mathcal{B})}(A) = A$.

Proposition 5.3 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then, $\overline{N_r(\mathcal{B})}(A + B) \subseteq (\overline{N_r(\mathcal{B})}(A) + \overline{N_r(\mathcal{B})}(B))$.*

Proof As ideals in a ring are normal subgroups, so the proof can be obtained as Proposition 4 in [1]. $\square$

Proposition 5.4 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then, $\overline{N_r(\mathcal{B})}(AB) \subseteq (\overline{N_r(\mathcal{B})}(A)(\overline{N_r(\mathcal{B})}(B)))$.*

Proof Let x be any element of $\overline{N_r(\mathcal{B})}(AB)$. Then, $x + \mathcal{B}_r \cap AB \neq \emptyset$ for some $\mathcal{B}_r \in P_r(B)$. Hence, there exists an element y in R such that $y \in x + \mathcal{B}_r$ and $y \in AB$. Thus, we can write $y = \sum_{i=1}^n a_i b_i$ for some $a_i \in A$ and $b_i \in B$. Now, we have $x \in y + \mathcal{B}_r = \sum_{i=1}^n a_i b_i + \mathcal{B}_r$. Since $\mathcal{B}_r$ is ideal of R , it follows that

$$x \in y + \mathcal{B}_r = \sum_{i=1}^n a_i b_i + \mathcal{B}_r = \sum_{i=1}^n (a_i + \mathcal{B}_r)(b_i + \mathcal{B}_r).$$

We have $x = \sum_{i=1}^n x_i y_i$ for some $x_i \in a_i + \mathcal{B}_r$ and $y_i \in b_i + \mathcal{B}_r$. Since a_i and x_i in the same class, and since $a_i \in A$ it follows that $a_i \in (x_i + \mathcal{B}_r) \cap A$ and similarly we have

$b_i \in (y_i + \underline{\mathcal{B}_r}) \cap B$. Now, we obtain $x_i \in \overline{N_r(\mathcal{B})}(A)$ and $y_i \in \overline{N_r(\mathcal{B})}(B)$ which yields that $x \in (\overline{N_r(\mathcal{B})}A)(\overline{N_r(\mathcal{B})}B)$. Therefore, $\overline{N_r(\mathcal{B})}(AB) \subseteq (\overline{N_r(\mathcal{B})}A)(\overline{N_r(\mathcal{B})}B)$. $\square$

Proposition 5.5 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then,*

$$(\overline{N_r(\mathcal{B})}A) + (\overline{N_r(\mathcal{B})}B) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (A + B).$$

Proof As ideals in a ring are normal subgroups, so the proof can be obtained as Proposition 5 in [1]. $\square$

Proposition 5.6 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then,*

$$(\overline{N_r(\mathcal{B})}A)(\overline{N_r(\mathcal{B})}B) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB).$$

Proof Suppose that x is an arbitrary element of $(\overline{N_r(\mathcal{B})}A)(\overline{N_r(\mathcal{B})}B)$. Then, $x = \sum_{i=1}^n a_i b_i$ for some $a_i \in (\overline{N_r(\mathcal{B})}A)$ and $b_i \in (\overline{N_r(\mathcal{B})}B)$. Thus, for some $\mathcal{B}r_i, \mathcal{B}r'_i \subseteq \mathcal{B}$, $a_i + \mathcal{B}r_i \cap A \neq \emptyset$ and $b_i + \mathcal{B}r'_i \cap B \neq \emptyset$. Hence, there exist $x_i \in a_i + \mathcal{B}r_i \cap A$ and $y_i \in b_i + \mathcal{B}r'_i \cap B$ for $1 \leq i \leq n$ and so $\sum_{i=1}^n x_i y_i \in AB$ and $\sum_{i=1}^n x_i y_i \in \sum_{i=1}^n (a_i + \mathcal{B}r_i)(b_i + \mathcal{B}r'_i) = \sum_{i=1}^n a_i b_i + \sum_{i=1}^n (\mathcal{B}r_i + \mathcal{B}r'_i) = x + \sum_{i=1}^n (\mathcal{B}r_i + \mathcal{B}r'_i)$. Thus, we can write $\sum_{i=1}^n x_i y_i \in x + \sum_{\mathcal{B}r_i \subseteq \mathcal{B}} \mathcal{B}r_i$. So that, $\sum_{i=1}^n x_i y_i \in x + \sum_{\mathcal{B}r \subseteq \mathcal{B}} \mathcal{B}r$. Hence, $x + \sum_{\mathcal{B}r \subseteq \mathcal{B}} \mathcal{B}r \cap AB \neq \emptyset$. Now we have $x \in \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB)$. Therefore, $(\overline{N_r(\mathcal{B})}A)(\overline{N_r(\mathcal{B})}B) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB)$. $\square$

Proposition 5.7 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then,*

$$(\underline{N_r(\mathcal{B})}A) + (\underline{N_r(\mathcal{B})}B) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (A + B).$$

Proof As ideals in a ring are normal subgroups, so the proof can be obtained as Proposition 6 in [1]. $\square$

Proposition 5.8 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A and B be two non-empty subsets of R . Then,*

$$(\underline{N_r(\mathcal{B})}A)(\underline{N_r(\mathcal{B})}B) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB).$$

Proof Suppose that x be an arbitrary element of $(\underline{N_r(\mathcal{B})A})(\underline{N_r(\mathcal{B})B})$. Then, $x = \sum_{i=1}^n a_i b_i$ for some $a_i \in (\underline{N_r(\mathcal{B})A})$ and $b_i \in (\underline{N_r(\mathcal{B})B})$. Thus, for some $\mathcal{B}r_i, \mathcal{B}r'_i \subseteq \mathcal{B}, a_i + \mathcal{B}r_i \subseteq A$ and $b_i + \mathcal{B}r'_i \subseteq B$. Since $\mathcal{B}r_i$ and $\mathcal{B}r'_i$ are ideals, it follows that $\sum_{i=1}^n (a_i + \mathcal{B}r_i)(b_i + \mathcal{B}r'_i) = \sum_{i=1}^n a_i b_i + \sum_{i=1}^n (\mathcal{B}r_i + \mathcal{B}r'_i) = x + \sum_{i=1}^n (\mathcal{B}r_i + \mathcal{B}r'_i) \subseteq AB$ which yields $x \in \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB)$. Therefore, $(\underline{N_r(\mathcal{B})A})(\underline{N_r(\mathcal{B})B}) \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB)$. $\square$

The following example shows that the converse of Proposition 5.8 is not true in general.

Example 8 Let $R = \mathbb{Z}_{12}, \mathcal{B} = \{I_1, I_2\}, I_1 = \{0, 4, 8\}, I_2 = \{0, 6\}, A = \{0, 2, 4, 6, 10\}$ and $B = \{0, 3, 6, 8, 9\}$. Then,

$$\begin{aligned} R/I_1 &= \{(0, 4, 8), (1, 5, 9), (2, 6, 10), (3, 7, 11)\}, \\ R/I_2 &= \{(0, 6), (1, 7), (2, 8), (3, 9), (4, 10), (5, 11)\}, \\ \underline{N_1(\mathcal{B})}(A) &= \{2, 6, 10\} \cup \{0, 6\} \cup \{4, 10\} = \{0, 2, 4, 6, 10\}, \\ \underline{N_1(\mathcal{B})}(B) &= \{0, 6\} \cup \{3, 9\} = \{0, 3, 6, 9\}, \\ (\underline{N_1(\mathcal{B})}A) \cdot (\underline{N_1(\mathcal{B})}B) &= \{0, 6\}, \\ A \cdot B &= \{0, 2, 4, 6, 8, 10\}, \\ \sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r &= \{I_1, I_2, I_1 + I_2\}, \\ I_1 + I_2 &= \{0, 2, 4, 6, 8, 10\}, \\ \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (AB) &= \{0, 2, 4, 6, 8, 10\}. \end{aligned}$$

Let I is an ideal of R and A non-empty subset of R then we define

$$I(A) = \{x : x \in I + a, a \in A\}.$$

Proposition 5.9 Let $(R, \mathbb{F})$ be a nearness approximation space. Let A be a non-empty subset of R . Then,

$$\overline{N_r(\mathcal{B})}A \subseteq \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r \right) (A).$$

Proof The proof can be obtained as Proposition 7 in [1]. $\square$

The inclusion symbol $\subseteq$ in Proposition 5.9 may not be replaced by an equals sign, as the next example shows.

Example 9 Let $R = \mathbb{Z}_{18}$, $\mathcal{B} = \{I_1, I_2, I_3\}$, $I_1 = \{0, 9\}$, $I_2 = \{0, 6, 12\}$, and $I_3 = \{0, 3, 6, 9, 12, 15\}$, $A = \{0, 1, 3, 5, 6, 9, 11, 13, 15\}$. Then, $I_{12} = \{0\}$, $I_{13} = \{0, 9\}$, $I_{23} = \{0, 6, 12\}$, $I_{12} + I_{13} = \{0, 9\} = I_1$, $I_{12} + I_{23} = \{0, 6, 12\} = I_2$, $I_{13} + I_{23} = \{0, 3, 6, 9, 12, 15\} = I_3$. Hence, by a simple calculation we have $\overline{N_2(\mathcal{B})A} = \mathbb{Z}_{18}/\{8, 16\}$ and $\left(\sum_{\mathcal{B}_2 \subseteq \mathcal{B}} \mathcal{B}_2\right)(A) = \mathbb{Z}_{18}$.

6 The lower and upper near subrings in a ring

In this section we introduce the notions of lower and upper near subrings on a near approximation space and study some of its properties.

Definition 6.1 Let $(R, \mathbb{F})$ be a nearness approximation space. Let A a non-empty subsets of R , then A is called an upper near subring (respectively, ideal) of R if $\overline{N_r(\mathcal{B})}(A)$ is a subring (respectively, ideal) of R . Similarly, A is called a lower near subring (respectively, ideal) of R if $\underline{N_r(\mathcal{B})}(A)$ is a subring (respectively, ideal) of R .

Proposition 6.2 Let $(R, \mathbb{F})$ be a nearness approximation space. Let A be a subring of R and $\overline{N_r(\mathcal{B})}(A) = \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r\right)(A)$. Then, A is an upper near subring of R .

Proof By Proposition 8 in [1], A is an upper near subgroup of R . Now, let a and b be two elements of $\overline{N_r(\mathcal{B})}(A)$. Then, there exist elements x and y in R such that $x \in a + \mathcal{B}_r \cap A$ and $y \in b + \mathcal{B}'_r \cap A$ for some $\mathcal{B}_r, \mathcal{B}'_r \subseteq \mathcal{B}$. Thus, $x, y \in A$, $x \in a + \mathcal{B}_r$, $y \in b + \mathcal{B}'_r$. Since A is a subring of R , it follows that $xy \in A$, and since $\mathcal{B}_r, \mathcal{B}'_r$ are ideals of R , then $xy \in (a + \mathcal{B}_r)(b + \mathcal{B}'_r) = ab + \mathcal{B}_r + \mathcal{B}'_r$. Thus, $xy \in ab + (\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r)$ and so that $xy \in ab + (\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r) \cap A$. Thus, $ab \in \overline{N_r(\mathcal{B})}(A)$ which yields that $\overline{N_r(\mathcal{B})}(A)$ is a subring of R . $\square$

Proposition 6.3 Let $(R, \mathbb{F})$ be a nearness approximation space. Let A be an ideal of R and $\overline{N_r(\mathcal{B})}(A) = \left(\sum_{\mathcal{B}_r \subseteq \mathcal{B}} \mathcal{B}_r\right)(A)$. Then, A is an upper near ideal of R .

Proof By Proposition 6.2, A is an upper near subring of R . Now, let a be any element of $\overline{N_r(\mathcal{B})}(A)$ and b be any element of R . Hence, there exists element x in R such that $x \in a + \mathcal{B}_r \cap A$. Thus, $x \in a + \mathcal{B}_r$ and $x \in A$. Since A is ideal of R , it follows that $rx \in A$. and since $\mathcal{B}_r$ is ideal of R , $rx \in r(a + \mathcal{B}_r) = ra + \mathcal{B}_r$, it follows that $rx \in ra + \mathcal{B}_r \cap A$. Thus, $ra \in \overline{N_r(\mathcal{B})}(A)$ so that $\overline{N_r(\mathcal{B})}(A)$ is an ideal of R . $\square$

Proposition 6.4 Let $(R, \mathbb{F})$ be a nearness approximation space. Let A be a subring of R and $\mathcal{B}_r \subseteq A$ for each $\mathcal{B}_r \subseteq \mathcal{B}$. Then, A is a lower near subring of R .

Proof By Proposition 10 in [1], A is an upper near subgroup of R . Now, let a and b be any element of $\underline{N_r(\mathcal{B})}(A)$. Thus, for some $\mathcal{B}_r, \mathcal{B}'_r \subseteq \mathcal{B}$, it yields $a + \mathcal{B}_r \subseteq A$ and $b + \mathcal{B}'_r \subseteq A$. Since $b = b + 0 \in A$, $\mathcal{B}_r \subseteq A$, and A is a subring of R then $b + \mathcal{B}_r \subseteq A$. Now since A is a subring of R and $\mathcal{B}_r$ is ideal of R , it follows that $ab + \mathcal{B}_r = (a + \mathcal{B}_r)(b + \mathcal{B}_r) \subseteq AA \subseteq A$. Hence, $ab \in \underline{N_r(\mathcal{B})}(A)$ which yields $\underline{N_r(\mathcal{B})}(A)$ is a subring of R . $\square$

Proposition 6.5 *Let $(R, \mathbb{F})$ be a nearness approximation space. Let A be an ideal of R and $\mathcal{B}_r \subseteq A$ for each $\mathcal{B}_r \subseteq \mathcal{B}$. Then, A is a lower near ideal of R .*

Proof By Proposition 6.4, A is a lower near subring of R . Now let a be any element of $\underline{N_r(\mathcal{B})}(A)$ and b be any element of R . Then, $a + \mathcal{B}_r \subseteq A$. Since A and $\mathcal{B}_r$ are ideals of R , it follows that $b(a + \mathcal{B}_r) = ba + \mathcal{B}_r \subseteq A$. This implies that $ba \in \underline{N_r(\mathcal{B})}(A)$ and so $\underline{N_r(\mathcal{B})}(A)$ is an ideal of R . $\square$

Proposition 6.6 *Let $(R, \mathbb{F})$ be a nearness approximation space. If I is an ideal of R and $\underline{Apr_B}(I) \neq \emptyset$, then $\underline{Apr_B}(I) = I$.*

Proof By Proposition 3.13 [4], $\underline{Apr_B}(I)$ is an ideal of R . Clearly, that $\underline{Apr_B}(I) \subseteq I$.

Conversely, let $a \in I$. Since $\underline{Apr_B}(I) \neq \emptyset$, it follows that there exists $x \in \underline{Apr_B}(I)$ such that $x + \mathcal{B} \subseteq I$. Since I is an ideal, $0 \in \mathcal{B}$, $x = x + 0 \in I$, it follows that $-x \in I$. Hence,

$$0 + \mathcal{B} = -x + x + \mathcal{B} \subseteq I + I = I.$$

This yields that $\mathcal{B} \subseteq I$. Since $a \in I$ and $\mathcal{B} \subseteq I$, it follows that $a + \mathcal{B} \subseteq I$. So that, $a \in \underline{Apr_B}(I)$. Therefore, $I \subseteq \underline{Apr_B}(I)$. $\square$

Let A be a non-empty subset of R and J is an ideal of R . Then, A is called an upper (respectively, lower) rough ideal of R if $\underline{Apr_J}(A)$ (respectively, $\underline{Apr_J}(A)$) is an ideal of R .

Proposition 6.7 *Let $(R, \mathbb{F})$ be a nearness approximation space. If I is an ideal of R and $\underline{Apr_B}(I) \neq \emptyset$, then $(\underline{Apr_B}(I), \underline{Apr_B}(I))$ is a rough ideal of R .*

Proof By Proposition 3.12 in [4], $\underline{Apr_B}(I)$ is an ideal of R . By Proposition 6.6 we have $\underline{Apr_B}(I) = I$. This implies that $\underline{Apr_B}(I)$ is an ideal of R . Therefore, $(\underline{Apr_B}(I), \underline{Apr_B}(I))$ is a rough ideal of R . $\square$

Let R be a commutative ring. An ideal $P \neq R$ is prime if and only if for any $a, b \in R$ and $ab \in P$ then $a \in P$ or $b \in P$. An ideal $P \neq R$ is primary if for any $a, b \in R$, $ab \in P$ and $a \notin P$ then $b^n \in P$ for some $n \geq 0$. Every prime ideal is clearly primary. if p is a prime integer and $n \geq 2$ a positive integer then $\langle p \rangle^n = \langle p^n \rangle$ is a primary ideal in $\mathbb{Z}$ which is not prime.

Proposition 6.8 *Let $(R, \mathbb{F})$ be a nearness approximation space. If I is a prime ideal of R and $\text{Apr}_{\mathcal{B}}(I) \neq \emptyset$, then I is a lower rough prime ideal of R .*

Proof The proof can be obtained as Proposition 6.6. $\square$

Proposition 6.9 *Let $(R, \mathbb{F})$ be a nearness approximation space. If I is a primary ideal and $\text{Apr}_{\mathcal{B}}(I) \neq \emptyset$, then I is a lower rough primary ideal of R .*

Proof The proof can be obtained as Proposition 6.6. $\square$

Proposition 6.10 *Let $(R, \mathbb{F})$ be a nearness approximation space and $\mathcal{B}, \mathcal{B}' \subseteq \mathbb{F}$ be ideals of R and A be a subring of R . Then,*

$$\begin{aligned}\overline{N_r(\mathcal{B})}(A) \cdot \overline{N_r(\mathcal{B}')}(A) &\subseteq \overline{N_r(\mathcal{B} + \mathcal{B}')}(A), \\ \underline{N_r(\mathcal{B})}(A) \cdot \underline{N_r(\mathcal{B}')}(A) &= \underline{N_r(\mathcal{B} + \mathcal{B}')}(A).\end{aligned}$$

Proof The proof can be obtained as Proposition 3.16 in [4]. $\square$

The inclusion symbol $\subseteq$ in Proposition 6.10 (i) may not be replaced by an equals sign, as the next example shows.

Example 10 Let $R = \mathbb{Z}_{21}$, $\mathcal{B} = \{I_1\}$, $I_1 = \{0, 7, 14\}$, $\mathcal{B}' = \{J_1\}$, $J_1 = \{0\}$ and $A = \{0, 3, 6, 9, 12, 15, 18\}$. Then, we have

$$\overline{N_1(\mathcal{B})}(A) = \text{Apr}_{I_1}(A) = \mathbb{Z}_{21} \quad \text{and} \quad \overline{N_1(\mathcal{B}')}(A) = \text{Apr}_{J_1}(A) = A.$$

Consequently, we obtain $\overline{N_1(\mathcal{B})}(A) \cdot \overline{N_1(\mathcal{B}')}(A) = \mathbb{Z}_{21} \cdot A = A$, whereas $\underline{N_1(\mathcal{B} + \mathcal{B}')}(A) = \mathbb{Z}_{21}$. This shows that the equality $\underline{N_r(\mathcal{B})}(A) \cdot \underline{N_r(\mathcal{B}')}(A) = \underline{N_r(\mathcal{B} + \mathcal{B}')}(A)$ is not true in general.

Proposition 6.11 *Let $(R, \mathbb{F})$ be a nearness approximation space and $\mathcal{B}, \mathcal{B}' \subseteq \mathbb{F}$ be ideals of R and A be a subring of R . Then, $\underline{N_r(\mathcal{B})}(A) + \underline{N_r(\mathcal{B}')}(A) = \underline{N_r(\mathcal{B} + \mathcal{B}')}(A)$.*

Proof The proof can be obtained as Proposition 3.18 in [4]. $\square$

Proposition 6.12 *Let $(R, \mathbb{F})$ be a nearness approximation space and $\mathcal{B}, \mathcal{B}' \subseteq \mathbb{F}$ be ideals of R and A be a subring of R . Then, $\overline{N_r(\mathcal{B})}(A) + \overline{N_r(\mathcal{B}')}(A) = \overline{N_r(\mathcal{B} + \mathcal{B}')}(A)$.*

Proof The proof can be obtained as Proposition 3.19 in [4]. $\square$

Proposition 6.13 *Let $(R, \mathbb{F})$ be a nearness approximation space and A be a subring of R . Then,*

$$\begin{aligned}\left(\overline{\sum_{B_r \subseteq B} B_r}\right)(A) &= \left(\sum_{B_r \subseteq B} \overline{B_r}\right)(A), \\ \left(\overline{\sum_{B_r \subseteq B} B_r}\right)(A) &= \left(\sum_{B_r \subseteq B} \overline{B_r}\right)(A).\end{aligned}$$

Proof The proof can be obtained as Proposition 12 in [1]. $\square$

Proposition 6.14 *Let $(R, \mathbb{F})$ be a nearness approximation space and A be a non-empty subset of R . Then, $\overline{N_r(\mathcal{B})}(A) = \left(\bigcup_{B_r \subseteq B} \overline{B_r}\right)(A)$.*

Proof The proof can be obtained as Proposition 14 in [1]. $\square$

Let R and R' be two rings. A function $f : R \rightarrow R'$ is called a homomorphism if $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$, for all $x, y \in R$. The kernel of f is define by $\text{Ker}f = \{x \in R : f(x) = 0\}$. Note that $\text{Ker}f$ is an ideal of R .

Proposition 6.15 *Let R and R' be two rings and f be a homomorphism from R to R' and A be a non-empty subset of R . Then,*

1. $f(\overline{N_r(\mathcal{B})}(A)) = \bigcup_{B_r \subseteq B} f(\overline{\text{Apr}_{B_r}}(A)),$
2. $f(\overline{N_r(\mathcal{B})}(A)) = \bigcup_{B_r \subseteq B} f(\overline{\text{Apr}_{B_r}}(A)),$
3. $f(\overline{(\mathcal{B} \cup \{\text{Ker}f\})}(A)) = f(\overline{\text{Apr}_{B_r}}(A)).$

Proof

1. This follow from Proposition 2 in [1] that $\overline{N_r(\mathcal{B})}(A) = \bigcup_{B_r \subseteq B} \overline{\text{Apr}_{B_r}}(A)$. Consequently, we have $f(\overline{N_r(\mathcal{B})}(A)) = f(\bigcup_{B_r \subseteq B} \overline{\text{Apr}_{B_r}}(A)) = \bigcup_{B_r \subseteq B} f(\overline{\text{Apr}_{B_r}}(A)).$
2. This follow from Proposition 2 in [1] that $\overline{N_r(\mathcal{B})}(A) = \bigcup_{B_r \subseteq B} \overline{\text{Apr}_{B_r}}(A)$. Hence, we have $f(\overline{N_r(\mathcal{B})}(A)) = f(\bigcup_{B_r \subseteq B} \overline{\text{Apr}_{B_r}}(A)) = \bigcup_{B_r \subseteq B} f(\overline{\text{Apr}_{B_r}}(A)).$
3. By Theorem 3.20 in [4], we have $f(\overline{\text{Apr}_{\text{Ker}f}}(A)) = f(A)$. Hence, we obtain $f(\overline{(\mathcal{B} \cup \{\text{Ker}f\})}(A)) = f(\overline{\text{Apr}_{B_r}}(A)) \cup f(\overline{\text{Apr}_{\text{Ker}f}}(A)) = f(\overline{\text{Apr}_{B_r}}(A)) \cup f(A) = f(\overline{\text{Apr}_{B_r}}(A)).$

$\square$

7 Compared approximations in a ring

In this section, the near approximations and the rough approximations are compared in a ring R . In addition basically, a nested chain of the near lower and near upper approximations is formed.

Proposition 7.1 *Let $(R, \mathbb{F})$ be a nearness approximation space and A be a non-empty subset of R .*

If $|\mathcal{B}| = r$ then, $\underline{N}_r(\mathcal{B})(A) = \underline{Apr}_{\mathcal{B}}(A)$ and $\overline{N}_r(\mathcal{B})(A) = \overline{Apr}_{\mathcal{B}}(A)$,

if $|\mathcal{B}| \geq 2$ then, $\underline{Apr}_{\mathcal{B}_i}(A) \subseteq \underline{N}_r(\mathcal{B})(A)$ and $\overline{Apr}_{\mathcal{B}_i}(A) \subseteq \overline{N}_r(\mathcal{B})(A)$.

Proof The proof can be obtained as Proposition 15 in [1]. $\square$

Proposition 7.2 *Let $(R, \mathbb{F})$ be a nearness approximation space and A be a non-empty subset of R . Let $\mathcal{B}_i \subseteq \mathcal{B}_j$ and $1 \leq i \leq j \leq |\mathcal{B}|$. Then,*

$$\underline{N}_r(\mathcal{B}_i)(A) \subseteq \underline{N}_r(\mathcal{B}_j)(A) \subseteq A \subseteq \overline{N}_r(\mathcal{B}_i)(A) \subseteq \overline{N}_r(\mathcal{B}_j)(A).$$

Proof The proof can be obtained as Proposition 16 in [1]. $\square$

Proposition 7.3 *Let $(R, \mathbb{F})$ be a nearness approximation space, A be a non-empty subset of R . Then, $\underline{N}_i(\mathcal{B})(A) \subseteq \underline{N}_j(\mathcal{B})(A) \subseteq A \subseteq \overline{N}_j(\mathcal{B})(A) \subseteq \overline{N}_i(\mathcal{B})(A)$, where $1 \leq i \leq j \leq |\mathcal{B}|$.*

Proof The proof can be obtained as Proposition 17 in [1]. $\square$

8 Conclusion

In this paper, we introduced an approach of ring theory considered in the context of near set theory. We demonstrated the concepts of near approximations in a ring are generalizations of the concepts of rough approximations in a ring [4]. We investigated some related properties about ideals, subrings, prime and primary ideals, and homomorphisms. Finally, we compared near approximations and rough approximations in a ring, and we deduced that rough approximations are special case of near approximations in rings.

In applied mathematics we encounter many examples of mathematical objects that can be added to each other and multiplied by scalar numbers. So, our future work on this argument will regard to modules over rings (or vector spaces). Indeed, near approximations in a vector space can be examined in future studies.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Human participants or animals This paper does not contain any studies with human participants or animals performed by any of the authors.

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